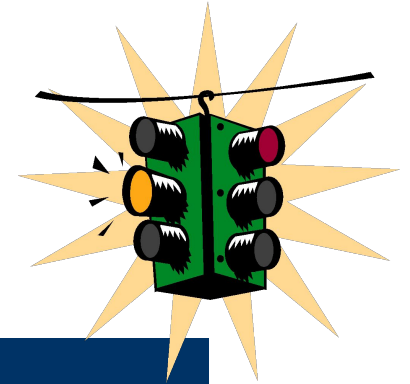


Input/Output and Communication



Introduction



I/O Processors

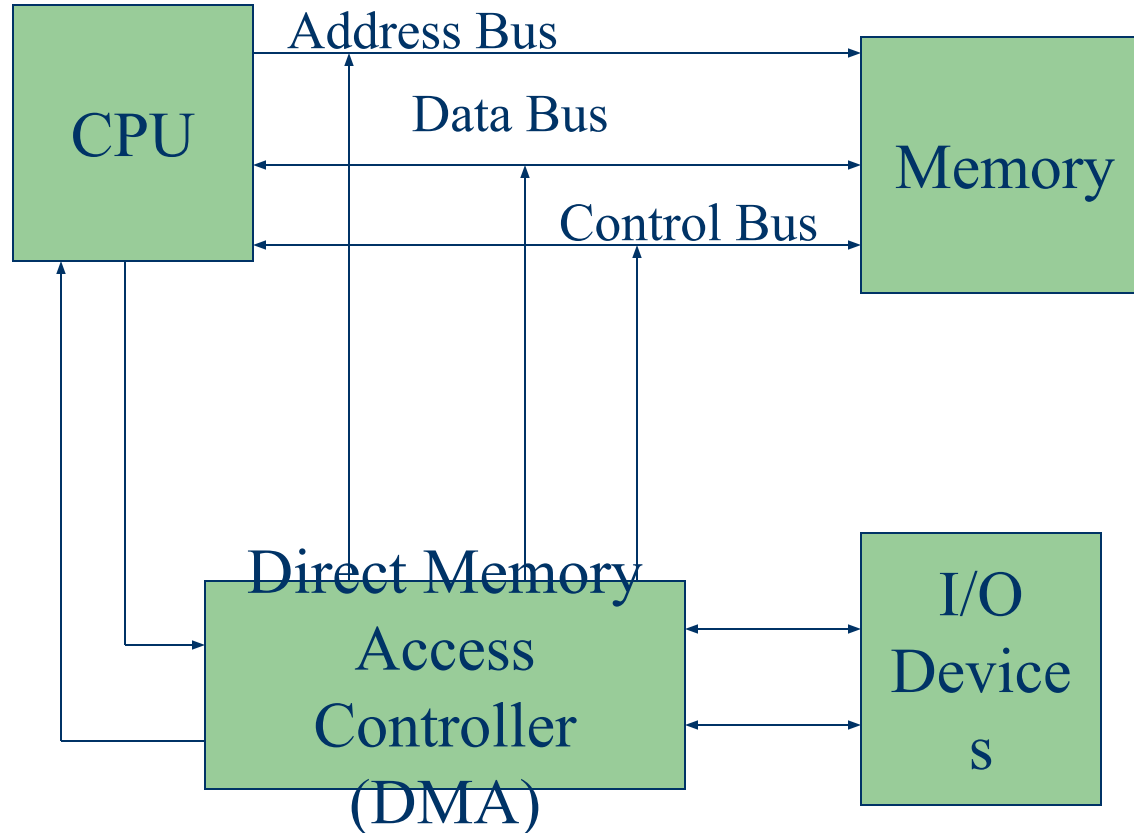
- Input-Output Processor
 - Review of DMA
 - Block transfer commands
 - ALU command
 - Control commands
- Serial Communication
 - Explanation of the Basic
 - Asynchronous serial communication
 - Synchronous transmission
 - Introduction of Universal Asynchronous Receiver Transmitter (UARTs)

Review of Direct Memory Access

DMA improves system performance by speeding up data transfer between memory and I/O System

- ❑ Bypass CPU, allow CPU to be use in another process.
- ❑ DMA controllers must manipulates each data transfer from I/O devices, and can only read.

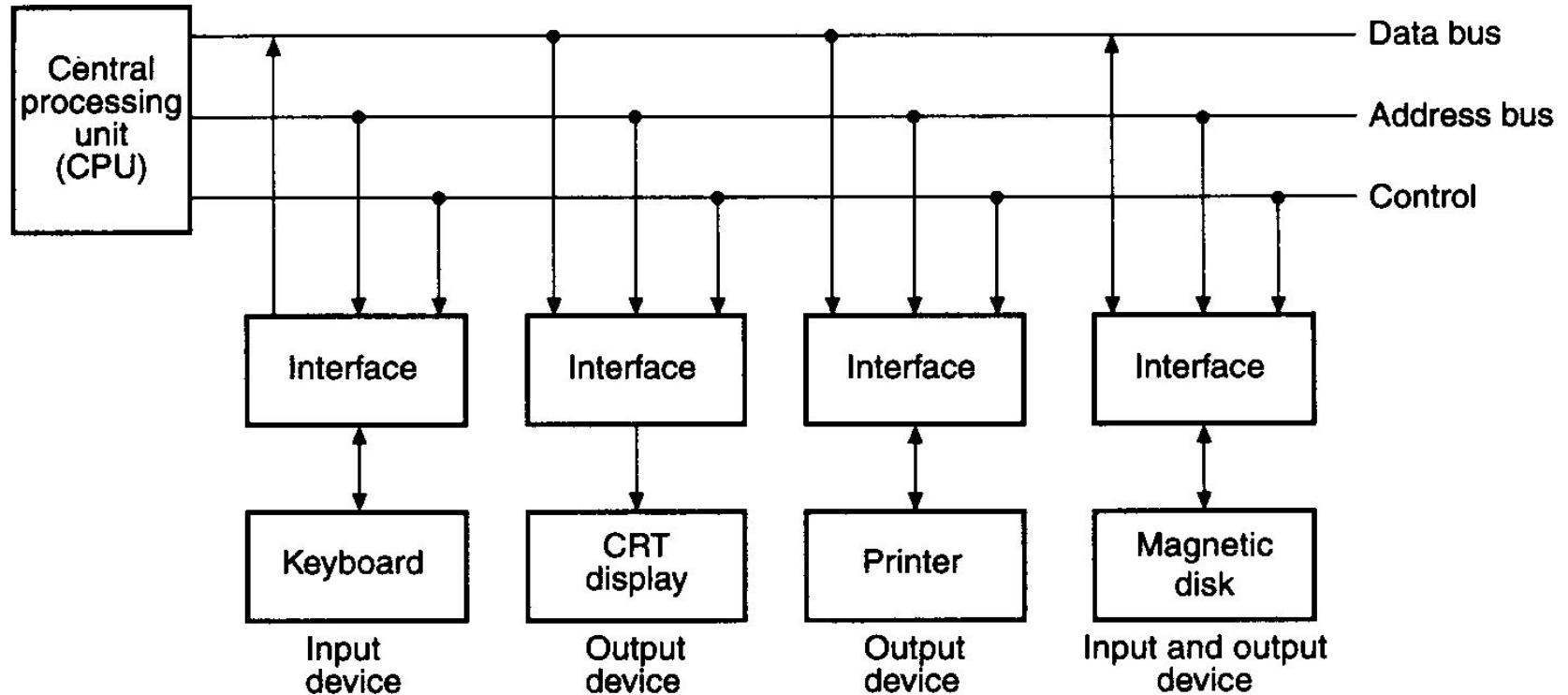
Direct Memory Access



I/O Processors (Introduction)

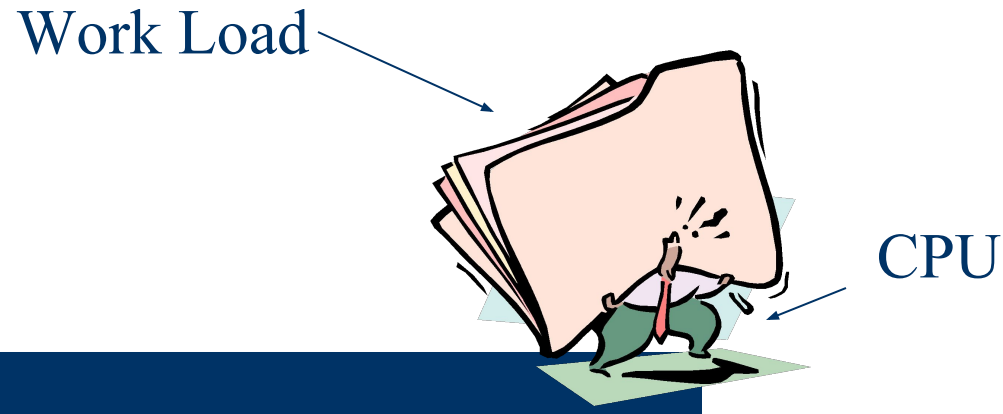
- ❑ I/O Processors also known as:
 - ❑ I/O Controllers
 - ❑ Channel Controllers
 - ❑ Peripheral Processing units (PPU)
 - ❑ Data Channel

CPU Connection to I/O Devices



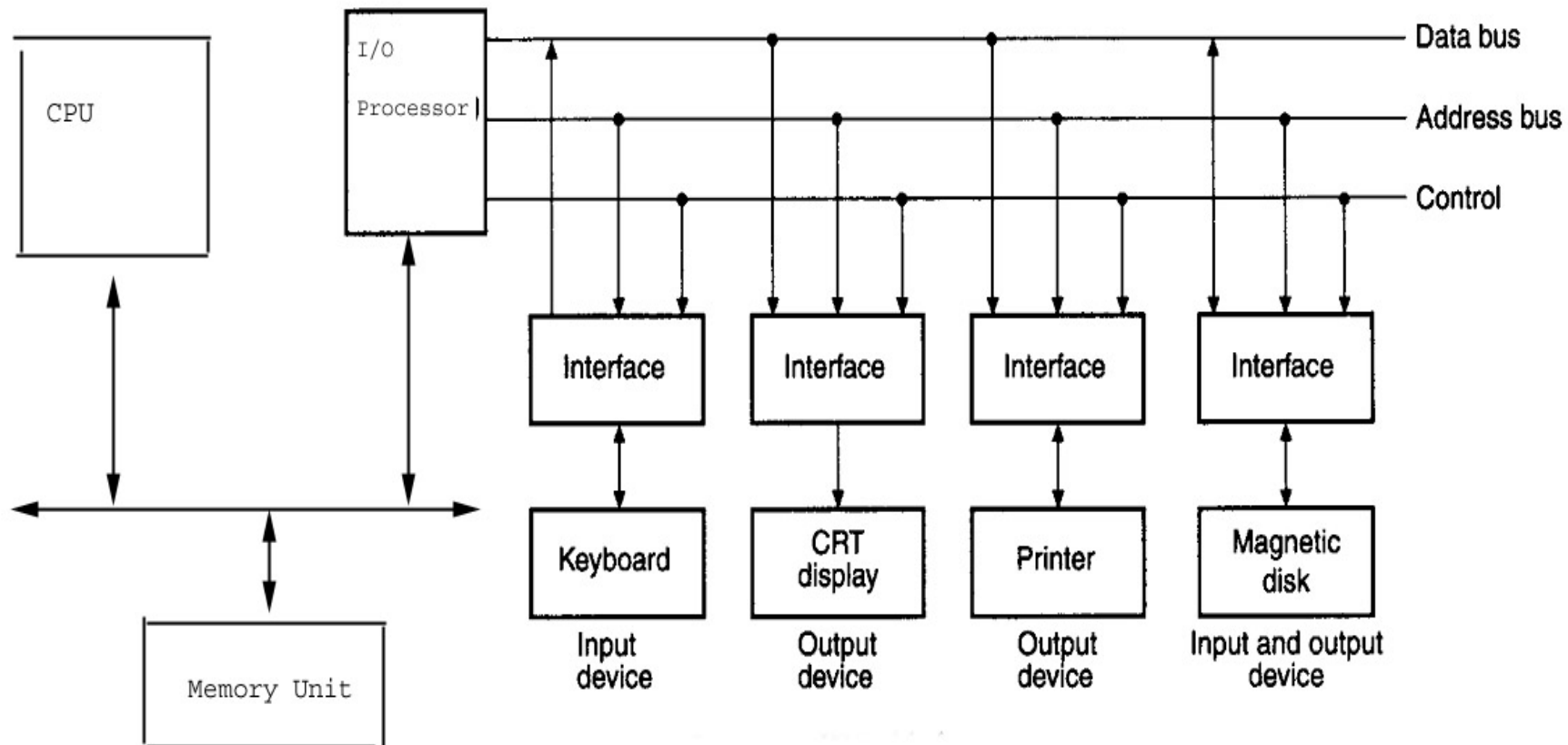
□ **FIGURE 11-4**
Connection of I/O Devices to CPU

I/O Processors



- ❑ I/O Processors handles all of the interactions between the I/O devices and the CPU.
- ❑ I/O Processors communicates with input and output devices through separate address, data, and control lines.
 - ❑ This provides an independent pathway for the transfer of information between external devices and internal memory.
- ❑ Relieves the CPU of 'I/O device chores'

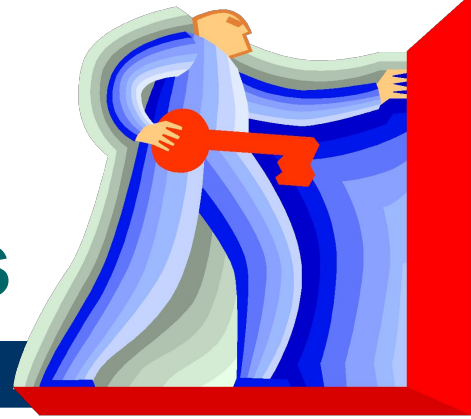
CPU with I/O Processor



TERMS

- Input-Output Processor (IOP)
 - Classified as a processor with direct memory access capability.
 - IOP fetches and execute its own instructions
 - Independent of the CPU
 - CPU initiating the IOP program
 - CPU is the master processor. IOP is considered the slave processors
 - There can be more than one or more IOP's

Three Forms of Commands



- Block transfer commands
 - Moves blocks data to IOP. Usually these instructions swap pages in and out of physical memory, and to load programs from disk memory.
- Arithmetic, logic, and Branch operations
 - IOP uses ALU instructions to manipulate the data so the process time for CPU is shorten.
- Control Command
 - Controls hardware.
 - Ex: rewind the tape on a tape drive or ejecting a CD from a drive.

The Process of the IOP



- CPU Instructions
 - Sends Command to test IOP path
 - Status approved and sends I/O commands
 - CPU continues with other process
 - Request IOP status
 - Check status for correct transfer.
- IOP Operations
 - Transfer status to Memory location
 - Access memory for IOP Commands
 - Conduct I/O transfer
 - I/O transfer completed, send status to CPU
 - Transfer status to memory location



Serial Communication



- Two forms of communication
 - Parallel communication
 - Transfers more than one bit of data at a given time
 - N-bits transmitted at the same time through n- wires
 - Faster but requires many wires and is used in short distances
 - EX: Input/output devices, DMA controllers, and I/O processors
 - Serial Communication
 - Serial communication refers to devices that cannot handle more than one bit of data at any given time by design.
 - Requires one wire and is slower.
 - Usually CPU use Parallel communication, if the device is serial, then the data is converted to use Parallel communication
 - EX: Modems

Two types of Serial Communication

- Asynchronous Serial Communication

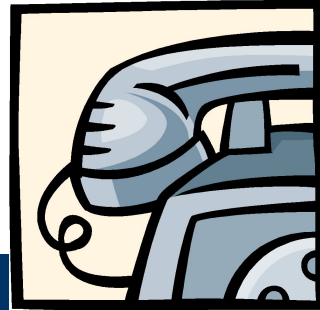
- Interacts with devices outside of the computer
 - Ex: modem connecting to another computer
- Transmit individual bytes instead of large blocks
- Do not share a common clock.

- Synchronous Serial transmission

- Transmits block of data in frames.
 - Frames are had head in front of the data and a tail at the end of the data.
- The head and tail contain information that allows the two computers to synchronize their clocks



Serial Communication Basics



- There are parameters that must be agreed upon between the two computer system. One of them is the speed.
 - Also known as Handshaking
 - Determine speed and transfer protocol
 - Speed is measured by the number of bits per second (bps)
 - Also Know as Baud Rate
- Device must agree on number of data bits per data transmission.
 - Parity Bits
 - Error checking
 - Stop Bits
 - End of transmission

Illustration of Handshaking

I am 14,400 bps

Ok

I am 28,800 bps

I will speak 14,000 for you

Data



Asynchronous Serial Communication

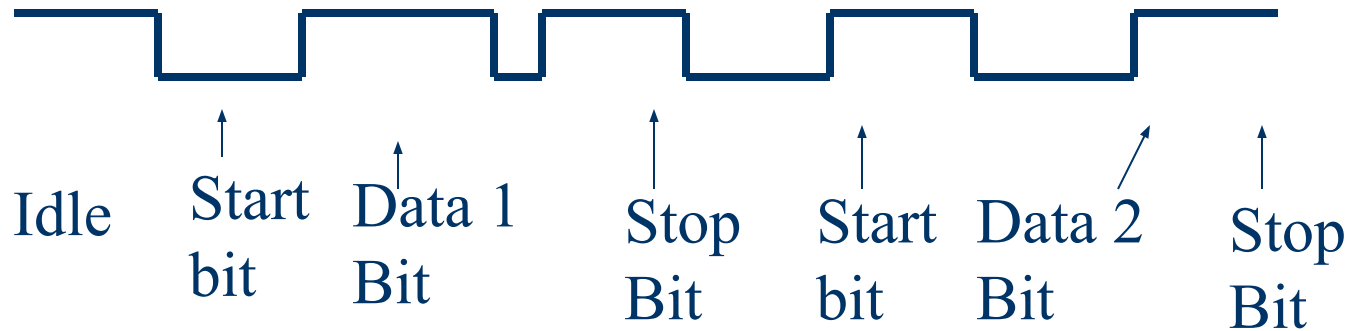


- Each byte is transmitted separate entity.
 - The Device must be able to recognize:
 - When transmission is occurring
 - When to read a bit of data
 - When the transmission ends
 - When the transmission is idle (no data being transmitted)

Asynchronous Serial Communication

- Device 1 transmission will output a 'start bit'
 - A line of transmission is used to describe the communication.
- Device 2 receives and confirms the bit.
- Device 2 begins to read a data bit off the line
- Then the process repeats however many data bits are on the line.
- The least significant bit is sent first and most important significant bit is sent last.
- Then Device 2 receives and confirms 'stop bit'

Line of transmission



Asynchronous Serial Communication

- Transmitting numeric data is straight forward
- Transmitting Characters are encoded with a binary value.
 - Most well know is American Standard Code for Information Interchange (ASCII)
 - Another is UNICODE.

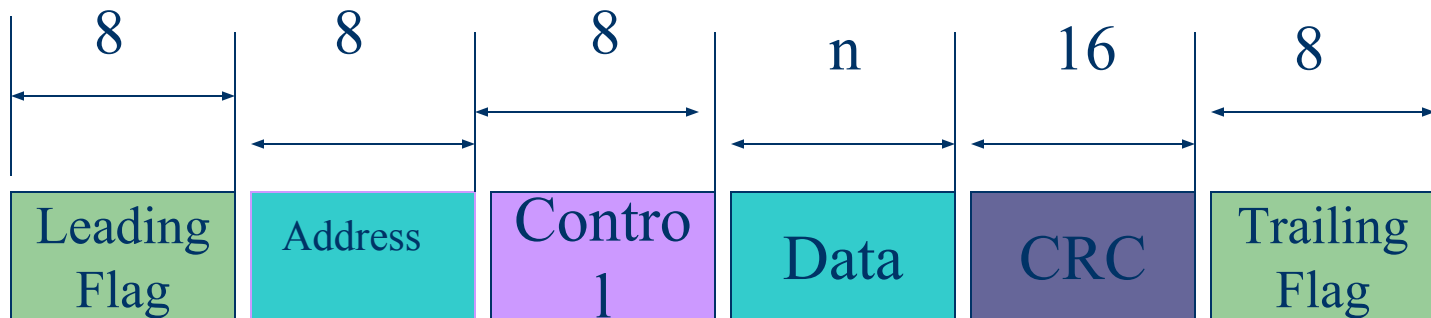


Synchronous Transmission

- Instead of transmitting a start and stop bit for each data value, Synchronous transmission strings together several data values into a data block called a 'frame.'
- There are several layers in the frame, similar to a data packet.
- There is a leading information, address of where the data is going, control ensure correct destination, the data it self, Cyclic redundancy check (CRC) to check there is no error and the trailing information

Synchronous Transmission Frame

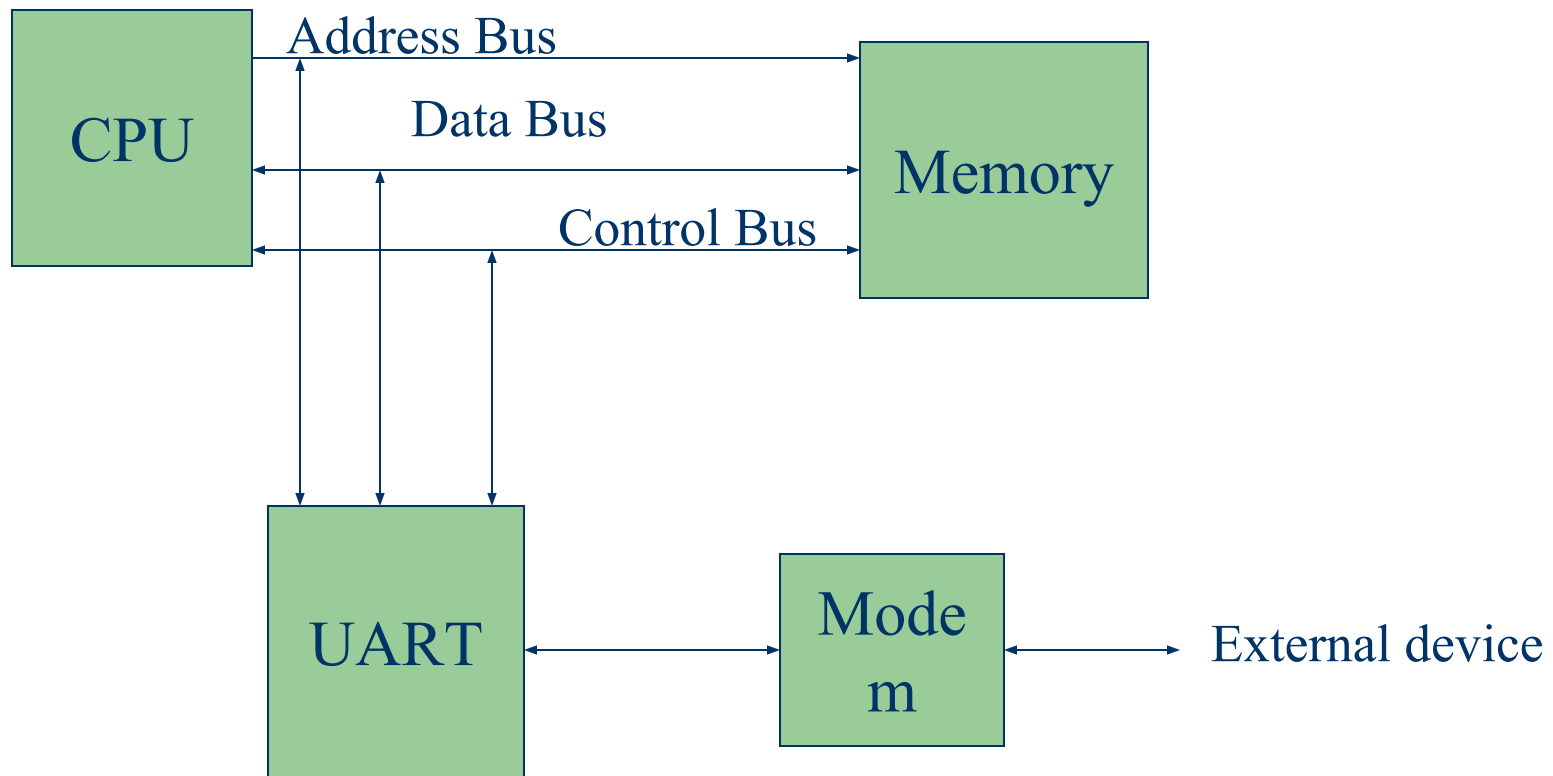
High Level Data Link Control
(HDLC)



Universal Asynchronous Receiver/Transmitters (UARTs)

- Asynchronous serial communication is a popular function. Manufacturers have designed special chip to deal with Asynchronous serial communication. This relieves the CPU of this task
- It is used to convert serial communication to parallel communication when receiving and convert parallel communication to serial when sending
- CPU sends command to the UARTs control register to determine the number of data bits, parity, and the number of stop bits to be used in transmission

UARTs



Conclusion

- Brief Review of Direct Memory Access
- Covered I/O Processor
 - Covered Input-Output Processor
 - Different commands for the I/O Processor
 - The Process
- Covered Serial Communication
 - Asynchronous Serial Communication
 - Synchronous Transmission
 - Universal Asynchronous Receiver/Transmitters (UARTs)
- Each topic showed ways to take the work load off of the CPU

